



α ALPHA PARTICLE

During alpha decay, an alpha particle (a helium nucleus comprising two protons and two neutrons) is emitted from the unstable parent nucleus.

γ GAMMA RAYS

High-energy photons are emitted during gamma decay.



SEEKING STABILITY

Radioactivity is the emission of waves or particles from an unstable atomic nucleus. This occurs when a nucleus spontaneously transforms into a more stable configuration itself by emitting energy. The random nature of the quantum world means it is impossible to predict when an individual nucleus will decay, although the decay of a large group of identical nuclei can be described by their half-life (time taken for half of the nuclei to decay).